



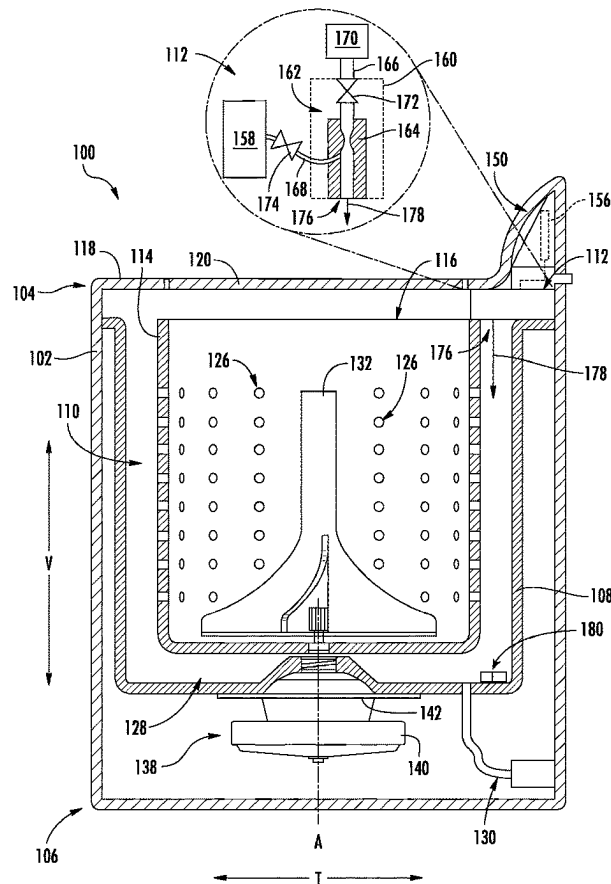
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Ewing(10) **Pub. No.: US 2025/0283263 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **METHOD OF SENSING LOAD IMBALANCES
IN A WASHING MACHINE APPLIANCE****G01M 1/16** (2006.01)**G01M 1/30** (2006.01)(71) Applicant: **Haier US Appliance Solutions, Inc.,**
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2105/58 (2020.02)(72) Inventor: **Cody Ewing**, Floyds Knobs, IN (US)(21) Appl. No.: **18/596,169**(22) Filed: **Mar. 5, 2024****Publication Classification**(51) **Int. Cl.****D06F 34/16** (2020.01)
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(57)

ABSTRACT

A washing machine appliance includes a drain pump fluidly coupled to a wash tub, a motor mechanically coupled to a wash basket, and a pressure sensor fluidly coupled to the wash tub. A controller is configured operate the motor to agitate a laundry load in the wash basket with a total volume of wash fluid in the wash tub, operate the drain pump to drain a portion of the total volume of the wash fluid from the wash tub until a remaining portion of the wash fluid remains in the wash tub, operate the motor to rotate the wash basket at a sensing speed, obtain a pressure signal from the pressure sensor, analyze the pressure signal to identify an out of balance condition, and implement a responsive action in response to identifying the out of balance condition.



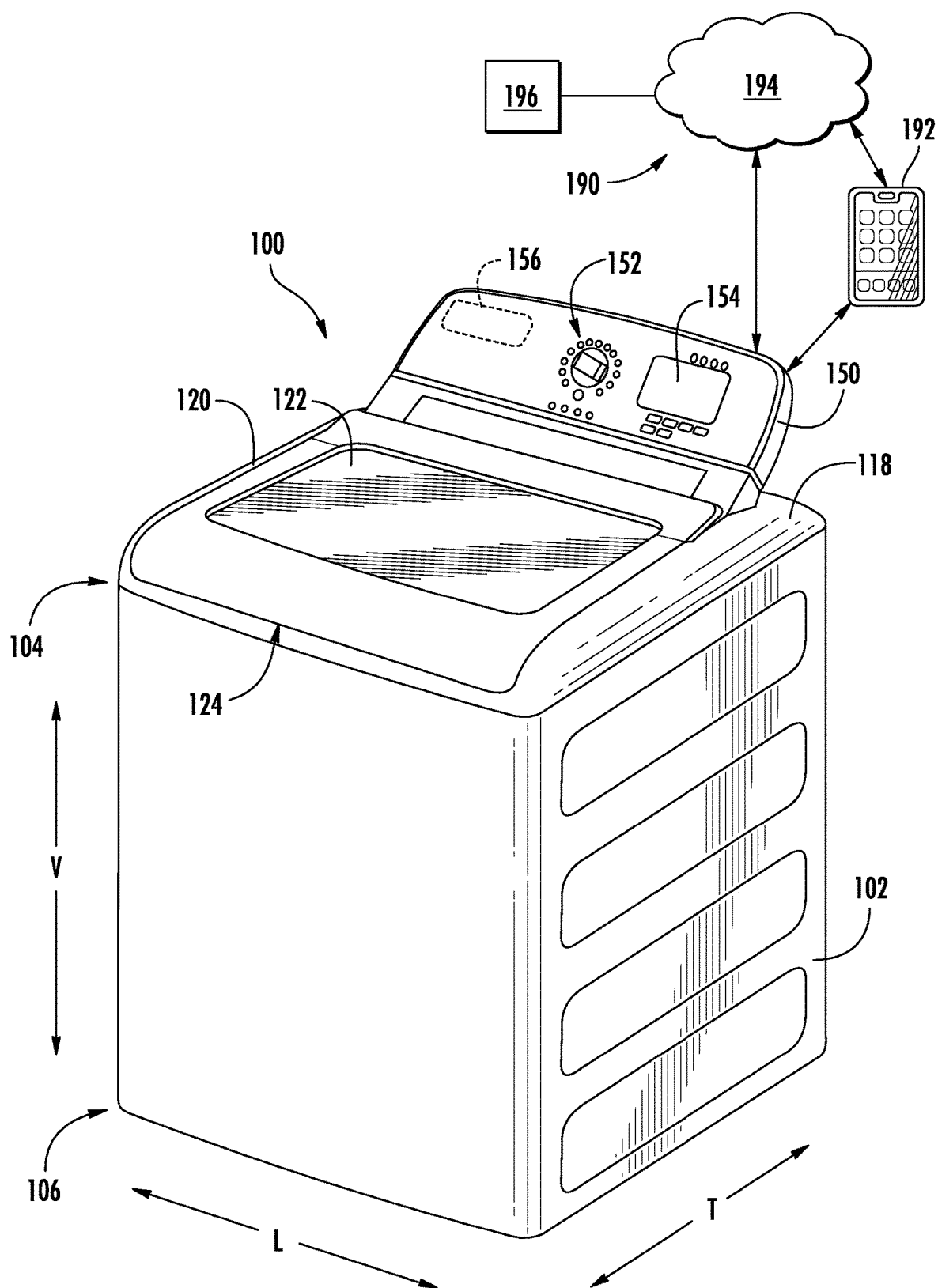


FIG. 1

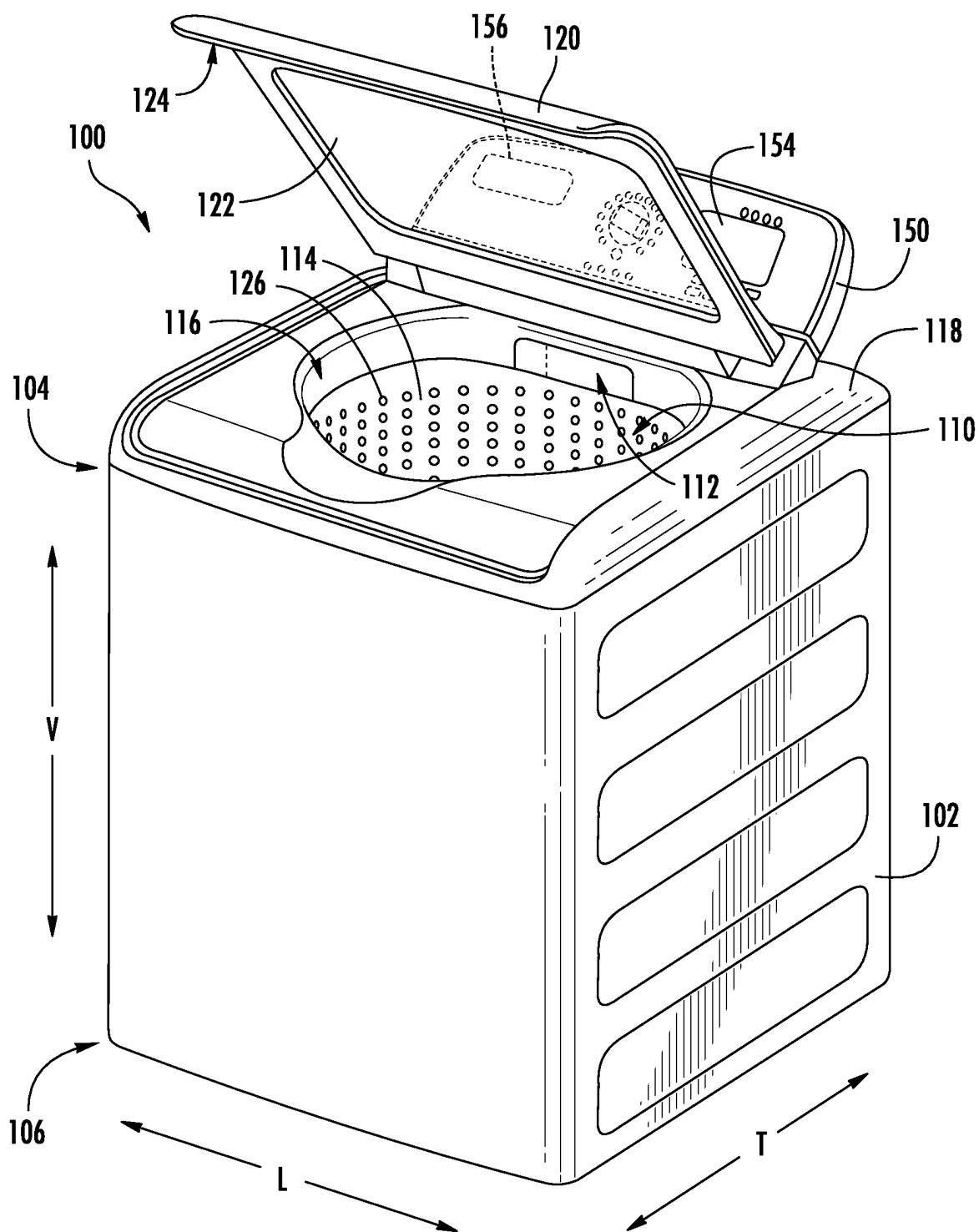


FIG. 2

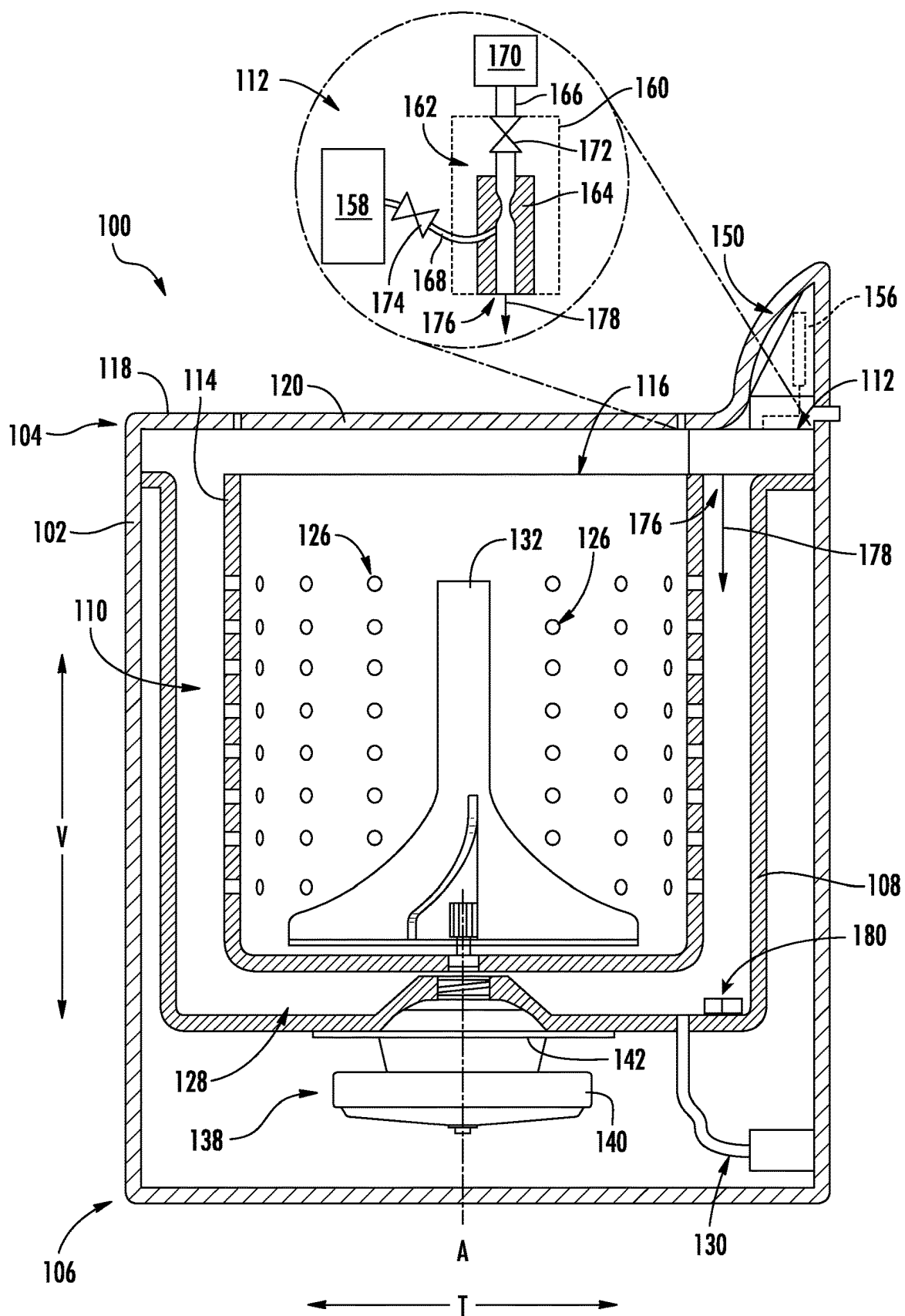


FIG. 3

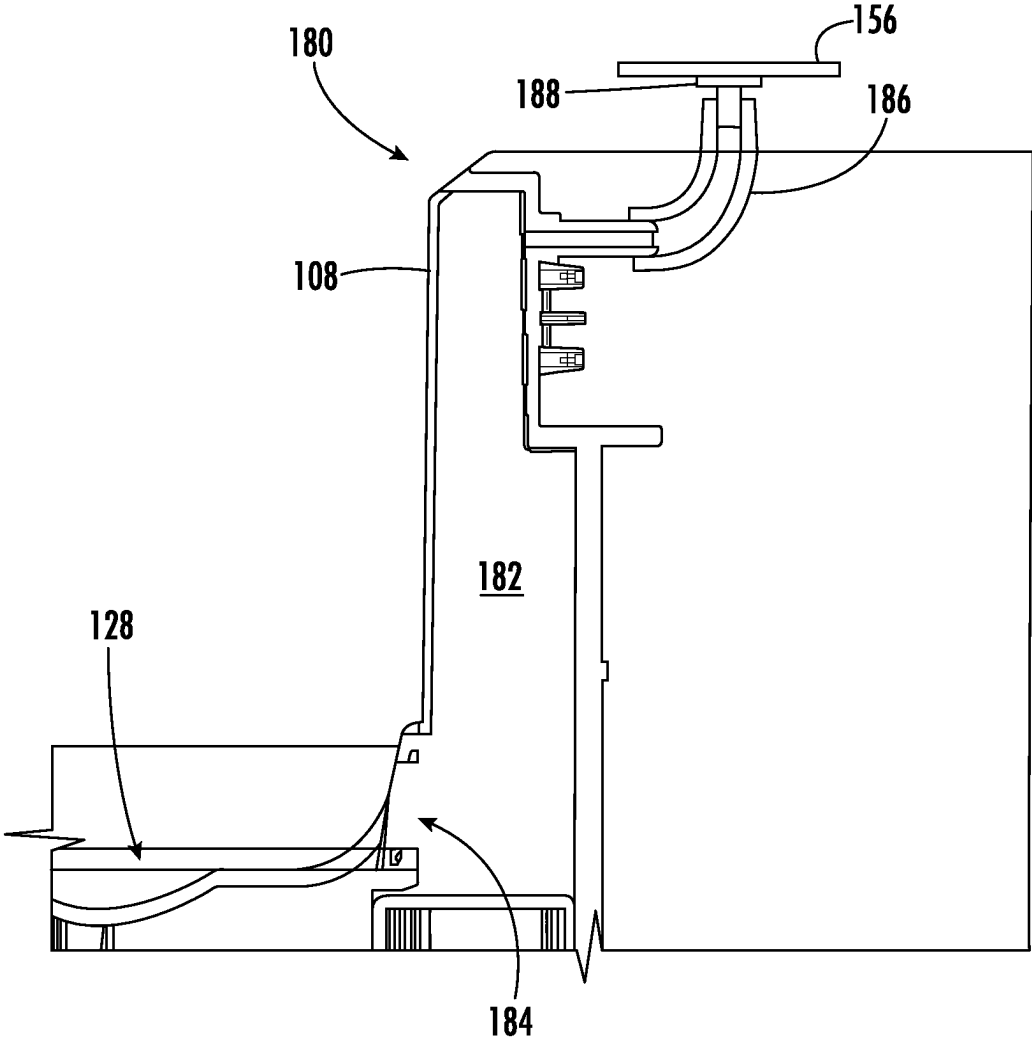


FIG. 4

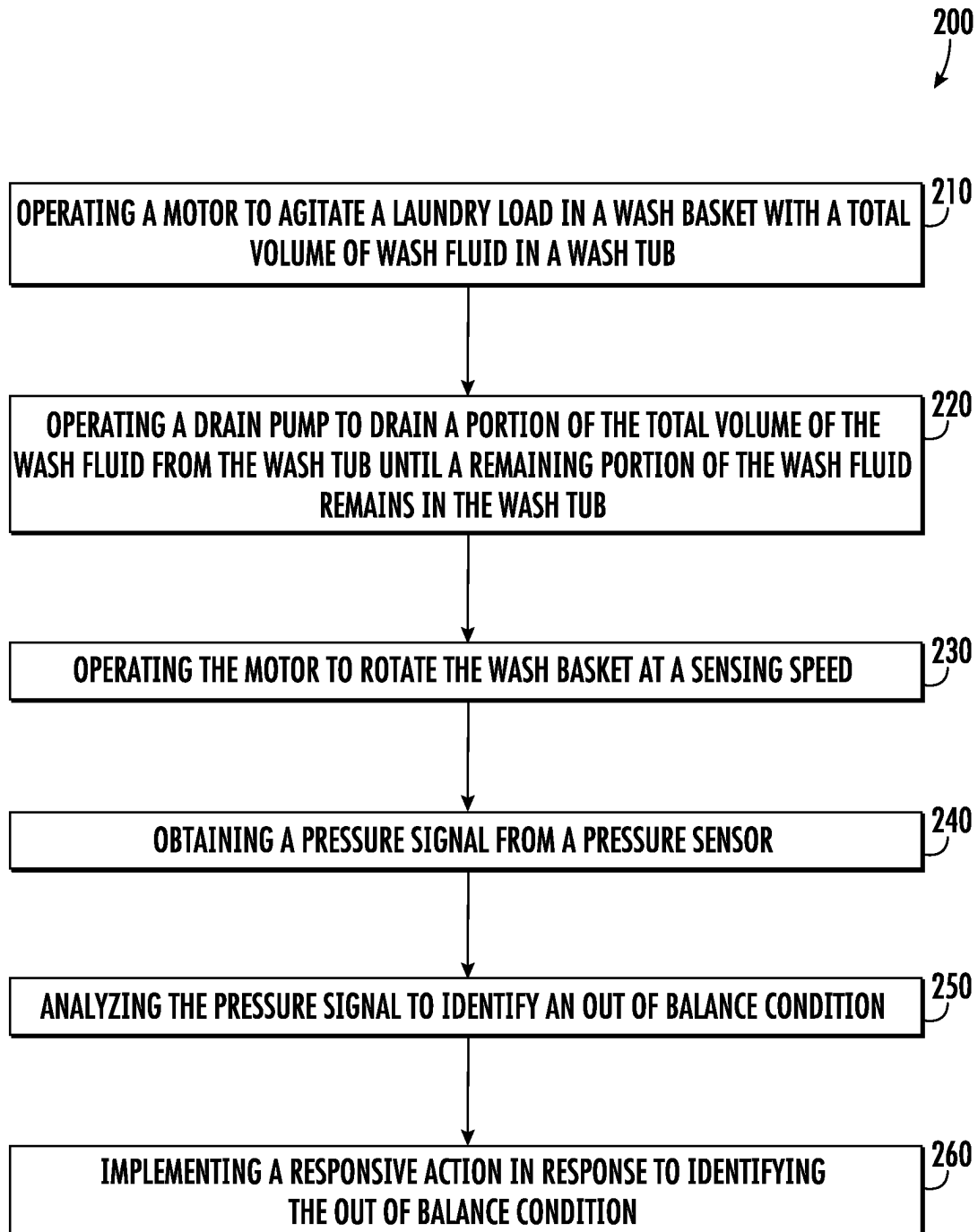


FIG. 5

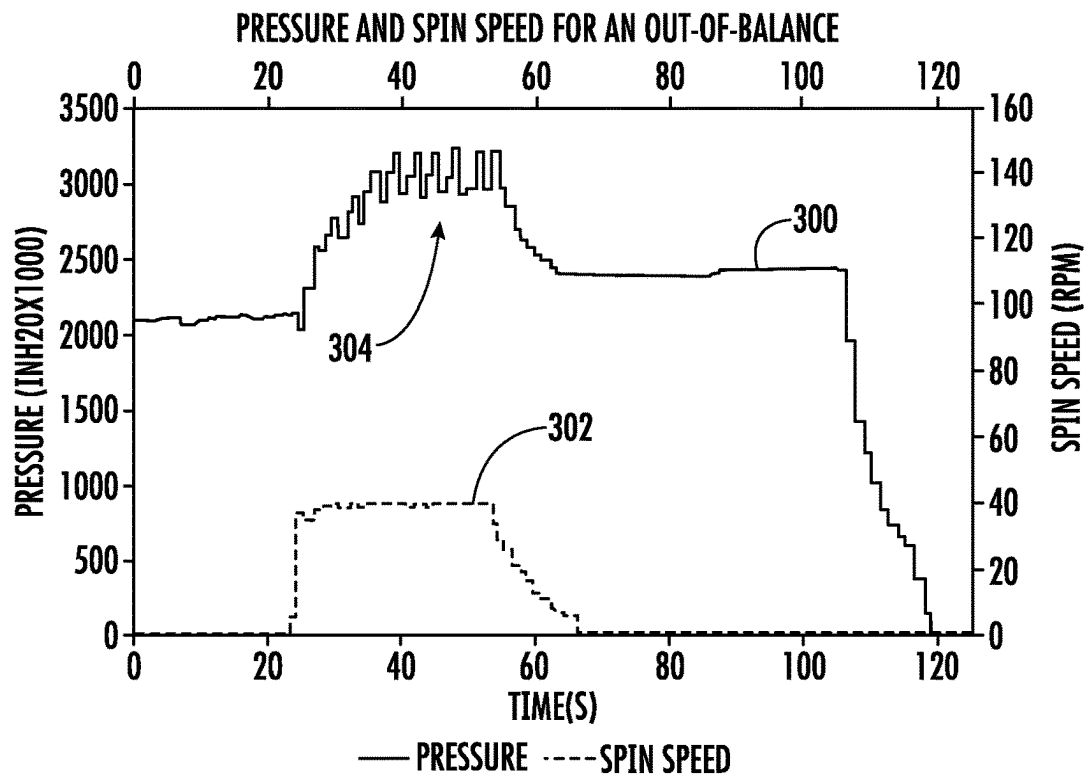


FIG. 6

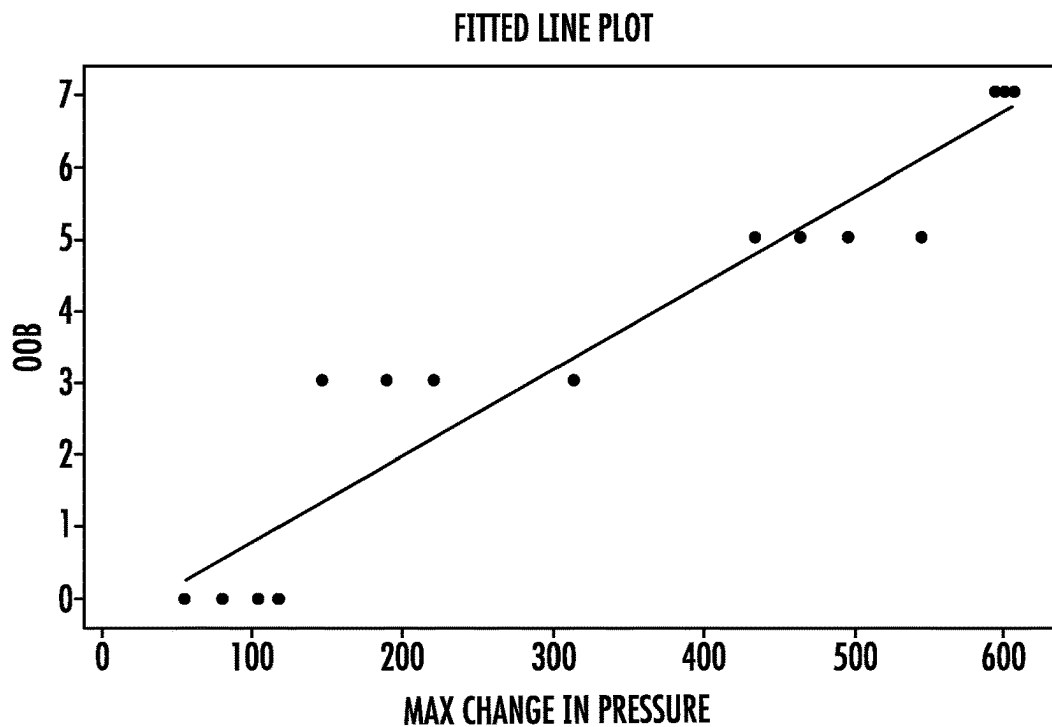


FIG. 7

METHOD OF SENSING LOAD IMBALANCES IN A WASHING MACHINE APPLIANCE

FIELD OF THE INVENTION

[0001] The present disclosure relates generally to top load washing machines, in particular a method of operating a top loading washing machine to sense a load imbalance prior to a spin cycle.

BACKGROUND OF THE INVENTION

[0002] Washing machine appliances generally include a cabinet which supports a wash tub for containing wash fluid, e.g., wash water, detergent, bleach, and/or other wash additives. A wash basket is mounted within the wash tub and defines a wash chamber for receipt of articles for washing. During operation of such washing machine appliances, wash fluid is directed into the wash tub and onto articles within the wash chamber of the wash basket. The wash basket or an agitation element can rotate at various speeds to agitate articles within the wash chamber in the wash fluid. A spin cycle is often included after some cycles, for example at the end of a wash cycle or a rinse cycle. During the spin cycle, the wash basket spins at a high speed to urge wash fluid from articles within the wash chamber.

[0003] In some cases, an uneven distribution of weight during a spin cycle in a top load washing machine may cause excessive displacement of the wash tub and wash basket. In minor cases, a slight imbalance may cause vibration or noise. In extreme cases, the wash tub may contact the cabinet of the appliance. These impacts or “cabinet strikes” are noisy and can cause the washing machine to “walk” on the floor. Over time, the impacts can cause damage to the appliance.

[0004] Some known top load washing machines sense a load imbalance during a spin cycle by sensing vibration or cabinet strikes. The excessive vibration/cabinet strikes that occur before the load imbalance is corrected can lead to consumer dissatisfaction or machine damage. Accordingly, a method of sensing an out of balance condition prior to initiating a spin cycle may be desirable.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, may be apparent from the description, or may be learned through practice of the invention.

[0006] In one exemplary embodiment, a washing machine appliance defining a vertical direction, a lateral direction, and a transverse direction is provided. The washing machine appliance includes a wash tub positioned within a cabinet, a drain pump fluidly coupled to the wash tub, a wash basket rotatably mounted within the wash tub, a motor mechanically coupled to the wash basket, a pressure sensor fluidly coupled to the wash tub, and a controller operably coupled to the drain pump, the motor, and the pressure sensor. The controller is configured to operate the motor to agitate a laundry load in the wash basket with a total volume of wash fluid in the wash tub, operate the drain pump to drain a portion of the total volume of the wash fluid from the wash tub until a remaining portion of the wash fluid remains in the wash tub, operate the motor to rotate the wash basket at a sensing speed, obtain a pressure signal from the pressure sensor, analyze the pressure signal to identify an out of

balance condition, and implement a responsive action in response to identifying the out of balance condition.

[0007] In another exemplary embodiment, a method of operating a washing machine appliance is provided, the washing machine appliance including a wash tub positioned within a cabinet, a drain pump fluidly coupled to the wash tub, a wash basket rotatably mounted within the wash tub, a motor mechanically coupled to the wash basket, and a pressure sensor fluidly coupled to the wash tub. The method includes operating the motor to agitate a laundry load in the wash basket with a total volume of wash fluid in the wash tub, operating the drain pump to drain a portion of the total volume of the wash fluid from the wash tub until a remaining portion of the wash fluid remains in the wash tub, operating the motor to rotate the wash basket at a sensing speed, obtaining a pressure signal from the pressure sensor, analyzing the pressure signal to identify an out of balance condition, and implementing a responsive action in response to identifying the out of balance condition.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 provides a perspective view of a washing machine appliance according to an example embodiment of the present subject matter with a lid in a closed position.

[0011] FIG. 2 provides a perspective view of the example washing machine appliance of FIG. 1 with the lid of the washing machine appliance shown in an open position according to an example embodiment of the present subject matter.

[0012] FIG. 3 provides a side cross-sectional view of the example washing machine appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0013] FIG. 4 provides a cross-sectional view of a pressure sensing port and a pressure sensor of the example washing machine appliance of FIG. 1 according to an example embodiment of the present subject matter.

[0014] FIG. 5 illustrates a method of determining an out of balance condition in a washing machine appliance according to an example embodiment of the present subject matter.

[0015] FIG. 6 is a plot of a sump pressure and basket spin speed according to an example embodiment of the present subject matter.

[0016] FIG. 7 is a plot of an out of balance magnitude relative to measured pressure changes according to an example embodiment of the present subject matter.

[0017] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0018] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0019] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0020] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

[0021] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one

embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0022] FIGS. 1 through 3 illustrate an exemplary embodiment of a vertical axis washing machine appliance 100. Specifically, FIGS. 1 and 2 illustrate perspective views of washing machine appliance 100 in a closed and an open position, respectively. FIG. 3 provides a side cross-sectional view of washing machine appliance 100. Washing machine appliance 100 generally defines a vertical direction V, a lateral direction L, and a transverse direction T, each of which is mutually perpendicular, such that an orthogonal coordinate system is generally defined.

[0023] While described in the context of a specific embodiment of vertical axis washing machine appliance 100, it should be appreciated that vertical axis washing machine appliance 100 is provided by way of example only. It will be understood that aspects of the present subject matter may be used in any other suitable washing machine appliance, such as a horizontal axis washing machine appliance. Indeed, modifications and variations may be made to washing machine appliance 100, including different configurations, different appearances, and/or different features while remaining within the scope of the present subject matter.

[0024] Washing machine appliance 100 has a cabinet 102 that extends between a top portion 104 and a bottom portion 106 along the vertical direction V, between a first side (left) and a second side (right) along the lateral direction L, and between a front and a rear along the transverse direction T. As best shown in FIG. 3, a wash tub 108 is positioned within cabinet 102, defines a wash chamber 110, and is generally configured for retaining wash fluids during an operating cycle. Washing machine appliance 100 further includes a primary dispenser or dispensing assembly 112 (FIG. 2) for dispensing wash fluid into wash tub 108.

[0025] In addition, washing machine appliance 100 includes a wash basket 114 that is positioned within wash tub 108 and generally defines an opening 116 for receipt of articles for washing. More specifically, wash basket 114 is rotatably mounted within wash tub 108 such that it is rotatable about an axis of rotation A. According to the illustrated embodiment, the axis of rotation A is substantially parallel to the vertical direction V. In this regard, washing machine appliance 100 is generally referred to as a “vertical axis” or “top load” washing machine appliance 100. However, it should be appreciated that aspects of the present subject matter may be used within the context of a horizontal axis or front load washing machine appliance as well.

[0026] As illustrated, cabinet 102 of washing machine appliance 100 has a top panel 118. Top panel 118 defines an opening (FIG. 2) that coincides with opening 116 of wash basket 114 to permit a user access to wash basket 114. Washing machine appliance 100 further includes a door 120 which is rotatably mounted to top panel 118 to permit selective access to opening 116. In particular, door 120 selectively rotates between the closed position (as shown in FIGS. 1 and 3) and the open position (as shown in FIG. 2). In the closed position, door 120 inhibits access to wash basket 114. Conversely, in the open position, a user can access wash basket 114. A window 122 in door 120 permits viewing of wash basket 114 when door 120 is in the closed

position, e.g., during operation of washing machine appliance **100**. Door **120** also includes a handle **124** that, e.g., a user may pull and/or lift when opening and closing door **120**. Further, although door **120** is illustrated as mounted to top panel **118**, door **120** may alternatively be mounted to cabinet **102** or any other suitable support.

[0027] As best shown in FIGS. 2 and 3, wash basket **114** further defines a plurality of perforations **126** to facilitate fluid communication between an interior of wash basket **114** and wash tub **108**. In this regard, wash basket **114** is spaced apart from wash tub **108** to define a space for wash fluid to escape wash chamber **110**. During a spin cycle, wash fluid within articles of clothing and within wash chamber **110** is urged through perforations **126** wherein it may collect in a sump **128** defined by wash tub **108**. Washing machine appliance **100** further includes a pump assembly **130** (FIG. 3) that is located beneath wash tub **108** and wash basket **114** for gravity assisted flow when draining wash tub **108**.

[0028] An impeller or agitation element **132** (FIG. 3), such as a vane agitator, impeller, auger, oscillatory basket mechanism, or some combination thereof is disposed in wash basket **114** to impart an oscillatory motion to articles and liquid in wash basket **114**. More specifically, agitation element **132** extends into wash basket **114** and assists agitation of articles disposed within wash basket **114** during operation of washing machine appliance **100**, e.g., to facilitate improved cleaning. In different embodiments, agitation element **132** includes a single action element (i.e., oscillatory only), a double action element (oscillatory movement at one end, single direction rotation at the other end) or a triple action element (oscillatory movement plus single direction rotation at one end, single direction rotation at the other end). As illustrated in FIG. 3, agitation element **132** and wash basket **114** are oriented to rotate about axis of rotation A (which is substantially parallel to vertical direction V).

[0029] As best illustrated in FIG. 3, washing machine appliance **100** includes a drive assembly or motor assembly **138** in mechanical communication with wash basket **114** to selectively rotate wash basket **114** (e.g., during an agitation or a rinse cycle of washing machine appliance **100**). In addition, motor assembly **138** may also be in mechanical communication with agitation element **132**. In this manner, motor assembly **138** may be configured for selectively rotating or oscillating wash basket **114** and/or agitation element **132** during various operating cycles of washing machine appliance **100**.

[0030] More specifically, motor assembly **138** may generally include one or more of a drive motor **140** and a transmission assembly **142**, e.g., such as a clutch assembly, for engaging and disengaging wash basket **114** and/or agitation element **132**. According to the illustrated embodiment, drive motor **140** is a brushless DC electric motor, e.g., a pancake motor. However, according to alternative embodiments, drive motor **140** may be any other suitable type or configuration of motor. For example, drive motor **140** may be an AC motor, an induction motor, a permanent magnet synchronous motor, or any other suitable type of motor. In addition, motor assembly **138** may include any other suitable number, types, and configurations of support bearings or drive mechanisms.

[0031] Referring still to FIGS. 1 through 3, a control panel **150** with at least one input selector **152** (FIG. 1) extends from top panel **118**. Control panel **150** and input selector **152** collectively form a user interface input for operator selection

of machine cycles and features. A display **154** of control panel **150** indicates selected features, operation mode, a countdown timer, and/or other items of interest to appliance users regarding operation.

[0032] Operation of washing machine appliance **100** is controlled by a controller or processing device **156** that is operatively coupled to control panel **150** for user manipulation to select washing machine cycles and features. In response to user manipulation of control panel **150**, controller **156** operates the various components of washing machine appliance **100** to execute selected machine cycles and features. According to an exemplary embodiment, controller **156** may include a memory and microprocessor, such as a general or special purpose microprocessor operable to execute programming instructions or micro-control code associated with methods described herein. Alternatively, controller **156** may be constructed without using a microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND gates, and the like) to perform control functionality instead of relying upon software. Control panel **150** and other components of washing machine appliance **100** may be in communication with controller **156** via one or more signal lines or shared communication busses.

[0033] During operation of washing machine appliance **100**, laundry items are loaded into wash basket **114** through opening **116**, and washing operation is initiated through operator manipulation of input selectors **152**. Wash basket **114** is filled with water and detergent and/or other fluid additives via primary dispenser **112**. One or more valves can be controlled by washing machine appliance **100** to provide for filling wash tub **108** and wash basket **114** to the appropriate level for the amount of articles being washed and/or rinsed. By way of example for a wash mode, once wash basket **114** is properly filled with fluid, the contents of wash basket **114** can be agitated (e.g., with agitation element **132** as discussed previously) for washing of laundry items in wash basket **114**.

[0034] After completion of the agitation cycle, washing machine appliance **100** may perform one or more rinse cycles. Specifically, according to an example embodiment, drain pump assembly **130** may drain the wash fluid from wash tub **108** and dispensing assembly **112** may dispense fresh water and/or a wash additive (such as fabric softener) into the wash tub. The load of clothes may then be agitated in the fresh water, e.g., to remove soil and detergent from load of clothes. After completion of the rinse cycle(s), drain pump assembly **130** may drain wash tub **108** and a spin cycle may be used to extract water from the clothes before the wash cycle is concluded.

[0035] Referring again to FIGS. 2 and 3, dispensing assembly **112** of washing machine appliance **100** will be described in more detail. As explained briefly above, dispensing assembly **112** may generally be configured to dispense wash fluid to facilitate one or more operating cycles or phases of an operating cycle (e.g., such as a wash cycle or a rinse cycle). The terms “wash fluid” and the like may be used herein to generally refer to a liquid used for washing and/or rinsing clothing or other articles. For example, the wash fluid is typically made up of water that may include other additives such as detergent, fabric softener, bleach, or other suitable treatments (including combinations thereof). More specifically, the wash fluid for a wash cycle may be a

mixture of water, detergent, and/or other additives, while the wash fluid for a rinse cycle may be water only and/or additional rinse additives.

[0036] As best shown schematically in FIG. 3, dispensing assembly 112 may generally include a bulk storage tank or bulk reservoir 158 and a dispenser box 160. More specifically, bulk reservoir 158 may be positioned under top panel 118 and defines an additive reservoir for receiving and storing wash additive. More specifically, according to the illustrated embodiment, bulk reservoir 158 may contain a bulk volume of wash additive (such as detergent or other suitable wash additives) that is sufficient for a plurality of wash cycles of washing machine appliance 100, such as no less than twenty wash cycles, no less than fifty wash cycles, etc. As a particular example, bulk reservoir 158 is configured for containing no less than twenty fluid ounces, no less than three-quarters of a gallon, or about one gallon of wash additive.

[0037] As will be described in detail below, dispensing assembly 112 may include features for drawing wash additive from bulk reservoir 158 and mixing it with water prior to directing the mixture into wash tub 108 to facilitate a cleaning operation. By contrast, dispensing assembly 112 is also capable of dispensing water only. Thus, dispensing assembly 112 may automatically dispense the desired amount of water with or without a desired amount of wash additive such that a user can avoid filling dispenser box 160 with detergent before each operation of washing machine appliance 100.

[0038] For example, as best shown in FIG. 3, washing machine appliance 100 includes an aspirator assembly 162, which is a Venturi-based dispensing system that uses a flow of water to create suction within a Venturi tube to draw in wash additive from bulk reservoir 158 which mixes with the water and is dispensed into wash tub 108 as a concentrated wash fluid preferably having a target volume of wash additive. After the target volume of wash additive is dispensed into wash tub 108, additional water may be provided into wash tub 108 as needed to fill to the desired wash volume. It should be appreciated that the target volume may be preprogrammed in controller 156 according to the selected operating cycle or parameters, may be set by a user, or may be determined in any other suitable manner.

[0039] As illustrated, aspirator assembly 162 includes a Venturi pump 164 that is fluidly coupled to both a water supply conduit 166 and a suction line 168. As illustrated, water supply conduit 166 may provide fluid communication between a water supply source 170 (such as a municipal water supply) and a water inlet of Venturi pump 164. In addition, washing machine appliance 100 includes a water fill valve or water control valve 172 which is operably coupled to water supply conduit 166 and is communicatively coupled to controller 156. In this manner, controller 156 may regulate the operation of water control valve 172 to regulate the amount of water that passes through aspirator assembly 162 and into wash tub 108.

[0040] In addition, suction line 168 may provide fluid communication between bulk reservoir 158 and Venturi pump 164 (e.g., via a suction port defined on Venturi pump 164). Notably, as a flow of water is supplied through Venturi pump 164 to wash tub 108, the flowing water creates a negative pressure within suction line 168. This negative pressure may draw in wash additive from bulk reservoir 158. When certain conditions exist, the amount of wash additive

dispensed is roughly proportional to the amount of time water is flowing through Venturi pump 164.

[0041] Referring still to FIG. 3, aspirator assembly 162 may further include a suction valve 174 that is operably coupled to suction line 168 to control the flow of wash additive through suction line 168 when desired. For example, suction valve 174 may be a solenoid valve that is communicatively coupled with controller 156. Controller 156 may selectively open and close suction valve 174 to allow wash additive to flow from bulk reservoir 158 through additive suction valve 174. For example, during a rinse cycle where only water is desired, suction valve 174 may be closed to prevent wash additive from being dispensed through suction valve 174. In some embodiments, suction valve 174 is selectively controlled based on at least one of the selected wash cycle, the soil level of the articles to be washed, and the article type. According to still other embodiments, no suction valve 174 is needed at all and alternative means for preventing the flow of wash additive may be used or other water regulating valves may be used to provide water into wash tub 108.

[0042] Washing machine appliance 100, or more particularly, dispensing assembly 112, generally includes a discharge nozzle 176 for directing a flow of wash fluid (e.g., identified herein generally by reference numeral 178) into wash tub 108. In this regard, discharge nozzle 176 may be positioned above wash tub 108 proximate a rear of opening 116 defined through top panel 118. Dispensing assembly 112 may be regulated by controller 156 to discharge wash fluid 178 through discharge nozzle 176 at the desired flow rates, volumes, and/or detergent concentrations to facilitate various operating cycles, e.g., such as wash or rinse cycles.

[0043] Although water supply conduit 166, water supply source 170, discharge nozzle 176, and water control valve 172 are all described and illustrated herein in the singular form, it should be appreciated that these terms may be used herein generally to describe a supply plumbing for providing hot and/or cold water into wash chamber 110. In this regard, water supply conduit 166 may include separate conduits for receiving hot and cold water, respectively. Similarly, water supply source 170 may include both hot- and cold-water supplies regulated by dedicated valves. In addition, washing machine appliance 100 may include one or more pressure sensors (not shown) for detecting the amount of water and/or clothes within wash tub 108. For example, the pressure sensor may be operably coupled to a side of wash tub 108 for detecting the weight of wash tub 108, which controller 156 may use to determine a volume of water in wash chamber 110 and a subwasher load weight.

[0044] After wash tub 108 is filled and the agitation phase of the wash cycle is completed, wash basket 114 can be drained, e.g., by drain pump assembly 130. Laundry articles can then be rinsed by again adding fluid to wash basket 114 depending on the specifics of the cleaning cycle selected by a user. The impeller or agitation element 132 may again provide agitation within wash basket 114. One or more spin cycles may also be used as part of the cleaning process. In particular, a spin cycle may be applied after the wash cycle and/or after the rinse cycle in order to wring wash fluid from the articles being washed. During a spin cycle, wash basket 114 is rotated at relatively high speeds to help wring fluid from the laundry articles through perforations 126. During or prior to the spin cycle, drain pump assembly 130 may operate to discharge wash fluid from wash tub 108, e.g., to

an external drain. After articles disposed in wash basket 114 are cleaned and/or washed, the user can remove the articles from wash basket 114, e.g., by reaching into wash basket 114 through opening 116.

[0045] Referring now specifically to FIGS. 3 and 4, washing machine appliance 100 may include a water level detection system 180 for determining the water level of wash fluid within wash tub 108. For example, water level detection system 180 may be a pressure-based sensing system that includes a pressure chamber 182 mounted to a bottom of wash tub 108, e.g., proximate sump 128. A chamber opening 184 may permit wash fluid to flow into pressure chamber until the wash fluid reaches a level that exceeds a top of chamber opening 184. As the water level of the wash fluid continues to rise, the air pressure within pressure chamber 182 rises accordingly, and the pressure rise may be used to determine the water level.

[0046] According to the illustrated embodiment, water level detection system 180 further includes a pressure hose 186 that seals off a port at a top of the pressure chamber 182. Pressure hose 186 extends upward to control panel 150, where it is fluidly coupled to a pressure sensor 188. In general, pressure sensor 188 may be any sensor suitable for determining a water level within sump 128 based on pressure readings. For example, pressure sensor 188 may be a piezoelectric pressure sensor and thus may include an elastically deformable plate and a piezoresistor mounted on the elastically deformable plate. According to exemplary embodiments, pressure sensor 188 is positioned proximate top 104 of cabinet 102, e.g., proximate or mounted to control panel 150. Thus, pressure hose 186 extends from pressure chamber 182 (i.e., proximate bottom 106 of cabinet 102) upward along the vertical direction V to pressure sensor 186.

[0047] Water level detection system 180 and pressure sensor 188 generally operate by measuring a pressure of air within pressure chamber 182 and using the measured chamber pressure to estimate the water level in sump 128. For example, when the water level within sump 128 falls below chamber opening 184, the pressure within pressure chamber 182 normalizes to ambient or atmospheric pressure, and thus reads a zero pressure. However, when water is present in sump 128 and rises above chamber opening 184, the measured air pressure becomes positive and may increase proportionally with the water level. Although sump 128 is described herein as containing water, it should be appreciated that aspects of the present subject matter may be used for detecting the level of any other suitable wash fluid.

[0048] Referring still to FIG. 1, a schematic diagram of an external communication system 190 will be described according to an exemplary embodiment of the present subject matter. In general, external communication system 190 is configured for permitting interaction, data transfer, and other communications between washing machine appliance 100 and one or more external devices. For example, this communication may be used to provide and receive operating parameters, user instructions or notifications, performance characteristics, user preferences, or any other suitable information for improved performance of washing machine appliance 100. In addition, it should be appreciated that external communication system 190 may be used to transfer data or other information to improve performance of one or more external devices or appliances and/or improve user interaction with such devices.

[0049] For example, external communication system 190 permits controller 156 of washing machine appliance 100 to communicate with a separate device external to washing machine appliance 100, referred to generally herein as an external device 192. As described in more detail below, these communications may be facilitated using a wired or wireless connection, such as via a network 194. In general, external device 192 may be any suitable device separate from washing machine appliance 100 that is configured to provide and/or receive communications, information, data, or commands from a user. In this regard, external device 192 may be, for example, a personal phone, a smartphone, a tablet, a laptop or personal computer, a wearable device, a smart home system, or another mobile or remote device.

[0050] In addition, a remote server 196 may be in communication with washing machine appliance 100 and/or external device 192 through network 194. In this regard, for example, remote server 196 may be a cloud-based server 196, and is thus located at a distant location, such as in a separate state, country, etc. According to an exemplary embodiment, external device 192 may communicate with a remote server 196 over network 194, such as the Internet, to transmit/receive data or information, provide user inputs, receive user notifications or instructions, interact with or control washing machine appliance 100, etc. In addition, external device 192 and remote server 196 may communicate with washing machine appliance 100 to communicate similar information.

[0051] In general, communication between washing machine appliance 100, external device 192, remote server 196, and/or other user devices or appliances may be carried using any type of wired or wireless connection and using any suitable type of communication network, non-limiting examples of which are provided below. For example, external device 192 may be in direct or indirect communication with washing machine appliance 100 through any suitable wired or wireless communication connections or interfaces, such as network 194. For example, network 194 may include one or more of a local area network (LAN), a wide area network (WAN), a personal area network (PAN), the Internet, a cellular network, any other suitable short- or long-range wireless networks, etc. In addition, communications may be transmitted using any suitable communications devices or protocols, such as via Wi-Fi®, Bluetooth®, Zigbee®, wireless radio, laser, infrared, Ethernet type devices and interfaces, etc. In addition, such communication may use a variety of communication protocols (e.g., TCP/IP, HTTP, SMTP, FTP), encodings or formats (e.g., HTML, XML), and/or protection schemes (e.g., VPN, secure HTTP, SSL).

[0052] External communication system 190 is described herein according to an exemplary embodiment of the present subject matter. However, it should be appreciated that the exemplary functions and configurations of external communication system 190 provided herein are used only as examples to facilitate description of aspects of the present subject matter. System configurations may vary, other communication devices may be used to communicate directly or indirectly with one or more associated appliances, other communication protocols and steps may be implemented, etc. These variations and modifications are contemplated as within the scope of the present subject matter.

[0053] While described in the context of a specific embodiment of vertical axis washing machine appliance

100, using the teachings disclosed herein it will be understood that vertical axis washing machine appliance **100** is provided by way of example only. Other washing machine appliances having different configurations, different appearances, and/or different features may also be utilized with the present subject matter as well, e.g., horizontal axis washing machine appliances. In addition, aspects of the present subject matter may be utilized in a combination washer/dryer appliance.

[0054] Now that the construction of washing machine appliance **100** and the configuration of controller **156** according to exemplary embodiments have been presented, an exemplary method **200** of operating a washing machine appliance will be described. Specifically, method **200** may be used to detect an out of balance condition in a washing machine appliance, such as the washing machine appliance **100**. Although the discussion below refers to the exemplary method **200** of operating washing machine appliance **100**, one skilled in the art will appreciate that the exemplary method **200** is applicable to the operation of a variety of other washing machine appliances, such as horizontal axis washing machine appliances. In exemplary embodiments, the various method steps as disclosed herein may be performed by controller **156** or a separate, dedicated controller.

[0055] Specifically, method **200** includes, at step **210**, operating a motor to agitate a laundry load in a wash basket with a total volume of wash fluid in the wash tub. In this regard, continuing the example from above, method **200** may be used to operate washing machine appliance **100** after a wash and/or agitation cycle where the load of clothes has been cleaned. In this regard, the load of clothes may be agitated within wash chamber **110** during the agitation cycle while the total volume of wash fluid is present, thereby cleaning the articles of clothing. As used herein, the “total volume of wash fluid” and the like are generally intended to refer to the target level of wash fluid associated with the agitation or wash cycle. In this regard, the total volume may be an amount (e.g., in gallons) needed to effectively wash the load of clothes during an agitation cycle.

[0056] Notably, a conventional wash cycle would enter directly into a spin and drain cycle after the agitation cycle. However, if the load of clothes is not evenly distributed within wash basket **114** at the end of the agitation cycle, immediately entering a high spin speed may result in undesirable vibrations due to the out of balance condition which may result in appliance damage, consumer dissatisfaction, etc. Accordingly, method **200** may provide a novel means for detecting out of balance conditions before entering the spin cycle. Specifically, step **220** may include operating a drain pump to drain a portion of the total volume of the wash fluid from the wash tub and until a remaining portion of the wash fluid remains in the wash tub. For example, drain pump assembly **130** may drain the wash fluid in wash tub **108** until a target volume (less than the total volume) is reached as determined by pressure sensor **188**. In this manner, the out of balance condition may be detected without causing extreme vibrations.

[0057] Step **230** may generally include operating the motor to rotate the wash basket at a sensing speed. In this regard, motor assembly **138** may rotate wash basket **114** while the remaining portion of wash fluid is present. For example, referring briefly to FIG. 6, a sump pressure **300** and a basket spin speed **302** are plotted over time when an out of balance condition is present. According to the illus-

trated embodiment, the sensing speed may be between about 10 and 70 (revolutions per minute) RPM, between about 30 and 50 RPM, or about 40 RPM. For example, the sensing speed may be lower than a spin speed or an extraction speed where wash fluid is actively extracted from the laundry load. By monitoring the sump pressure while the basket is rotated at the sensing speed with the remaining portion of wash fluid, and out of balance condition may be detected prior to initiating a spin cycle.

[0058] Specifically, step **240** may further include obtaining a pressure signal from a pressure sensor. In this regard, water level detection assembly **180** and pressure sensor **188** may communicate to controller **156** a sump pressure. This pressure signal may be obtained periodically or continuously and may be used to determine the water level in wash tub **108**. Step **250** may generally include analyzing the pressure signal to identify an out of balance condition. In this regard, an out of balance condition is illustrated in the region **304** of sump pressure **300**. Although exemplary methods of analyzing the pressure signal are described below, it should be appreciated that other methods and algorithms may be used while remaining within the scope of the present subject matter.

[0059] For example, analyzing the pressure signal to identify the out of balance condition may include quantifying the oscillations of the detected out of balance condition **304**. For example, a maximum pressure and a minimum pressure may be identified from the pressure signal and an oscillation amplitude (e.g., the difference between the maximum and minimum pressures) may be determined. An out of balance condition may be identified when the oscillation amplitude exceeds a predetermined amplitude threshold. For example, this predetermined amplitude threshold may be predetermined and set by the manufacturer based on empirical testing or theoretical data. According to still other embodiments, analyzing the pressure signal may include determining a running average of the pressure signal **300** and determining that the maximum or minimum pressures differ from the running average by more than a predetermined amount. As shown for example in FIG. 7, larger out of balance conditions may result in larger oscillations in sump pressure, and step **250** may include monitoring these oscillations to provide insight as to the severity of the out of balance condition.

[0060] Step **260** may generally include implementing a responsive action in response to identifying the out of balance condition. For example, implementing the responsive action may include performing a load redistribution procedure. For example, the load redistribution may include adding wash fluid to wash tub **108** and re-agitating the load of clothes to improve the distribution. According to example embodiments, steps **220** through **260** may be performed again until the out of balance condition is cured. In addition, implementing the responsive action in response to identifying the out of balance condition may further include providing a user notification of the out of balance condition. For example, this user notification may be provided through a control panel **150**, e.g., via display **154**. According to still other embodiments, the user notification may be provided to a remote device **192** (e.g., such as a user's cell phone) using network **194**. This user notification may include recommendations on rebalancing the load of clothes or other instructions for rectifying the out of balance condition.

[0061] Method 200 may further include proceeding with a spin cycle if an out of balance condition is not detected. Accordingly, method 200 may include analyzing the pressure signal to determine that the out of balance condition does not exist (e.g., corresponding to low oscillations in sump pressure 300). If the out of balance condition does not exist, method 200 may further include performing a spin cycle and completing the operating cycle of washing machine appliance 100.

[0062] FIG. 5 depicts steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that the steps of any of the methods discussed herein can be adapted, rearranged, expanded, omitted, or modified in various ways without deviating from the scope of the present disclosure. Moreover, although aspects of method 200 are explained using washing machine appliance 100 as an example, it should be appreciated that this method may be applied to the operation of any suitable laundry appliance, such as another washing machine appliance.

[0063] As explained herein, aspects of the present subject matter are generally directed to a method for sensing load unbalance using a piezoelectric pressure sensor in a washing machine. The method may include the performance of a unique spin step with some amount of water still in a tub and a basket, i.e., water is drained to a specified pressure sensor reading and the basket is spun at 40 RPM. An existing pressure sensor at the base of the tub may be utilized to take periodic measurements as the basket is spinning. The sensor may report a change in output voltage when a mechanical stress is applied to it, i.e., increase in air pressure; water level increases/decreases in the tub, which causes the pressure to increase/decrease at the pressure sensor since it is connected to the tub with a hose. Software then converts the analog voltage output from the sensor to a digital signal, and a mathematical formula may be used to convert the digital signal into a real-world pressure value. The water in the tub will gather in the tub opposite or with the unbalanced load (depending on speed), resulting in a “wave” that passes over the pressure sensor. This “wave” causes the voltage value coming from the pressure sensor to go up and down; this is then converted to pressure amplitude that is used to estimate the unbalanced load size. The pressure amplitude may be calculated by subtracting the running average pressure from the maximum peak pressure, or by subtracting the minimum peak pressure from the maximum peak pressure. Once the pressure amplitude is determined, the washing machine uses logic to determine whether the load should be redistributed or go into the spin dry step. The detection method prevents noisy cabinet impacts and high vibrations.

[0064] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

What is claimed is:

1. A washing machine appliance defining a vertical direction, a lateral direction, and a transverse direction, the washing machine appliance comprising:

a wash tub positioned within a cabinet;
a drain pump fluidly coupled to the wash tub;
a wash basket rotatably mounted within the wash tub;
a motor mechanically coupled to the wash basket;
a pressure sensor fluidly coupled to the wash tub; and
a controller operably coupled to the drain pump, the motor, and the pressure sensor, the controller being configured to:

operate the motor to agitate a laundry load in the wash basket with a total volume of wash fluid in the wash tub;

operate the drain pump to drain a portion of the total volume of the wash fluid from the wash tub until a remaining portion of the wash fluid remains in the wash tub;

operate the motor to rotate the wash basket at a sensing speed;

obtain a pressure signal from the pressure sensor;
analyze the pressure signal to identify an out of balance condition; and

implement a responsive action in response to identifying the out of balance condition.

2. The washing machine appliance of claim 1, wherein analyzing the pressure signal to identify the out of balance condition comprises:

identifying a maximum pressure and a minimum pressure from the pressure signal;

determining an oscillation amplitude of the pressure signal from the maximum pressure and the minimum pressure; and

identifying the out of balance condition when the oscillation amplitude exceeds a predetermined amplitude threshold.

3. The washing machine appliance of claim 1, wherein the sensing speed is lower than an extraction speed where the wash fluid is extracted from the laundry load.

4. The washing machine appliance of claim 1, wherein the sensing speed is less than 50 revolutions per minute.

5. The washing machine appliance of claim 1, wherein the analysis to identify the out of balance condition is performed before a spin cycle.

6. The washing machine appliance of claim 1, wherein the controller is further configured to:

analyze the pressure signal to determine the out of balance condition does not exist; and

perform a spin cycle in response to determining the out of balance condition does not exist.

7. The washing machine appliance of claim 1, wherein implementing the responsive action comprises:
performing a load redistribution procedure.

8. The washing machine appliance of claim 1, wherein implementing the responsive action comprises:
providing a user notification of the out of balance condition.

9. The washing machine appliance of claim 8, wherein the user notification is provided through a user interface panel.

10. The washing machine appliance of claim 8, wherein the controller is in operative communication with a remote device through an external network, and wherein the user notification is provided through the remote device.

11. The washing machine appliance of claim **1**, wherein the pressure sensor is a piezoelectric pressure sensor that is positioned at a user interface panel and is fluidly coupled to a pressure port at a lower portion of the wash tub.

12. The washing machine appliance of claim **1**, wherein the washing machine appliance is a top load washing machine appliance.

13. A method of operating a washing machine appliance, the washing machine appliance comprising a wash tub positioned within a cabinet, a drain pump fluidly coupled to the wash tub, a wash basket rotatably mounted within the wash tub, a motor mechanically coupled to the wash basket, and a pressure sensor fluidly coupled to the wash tub, the method comprising:

operating the motor to agitate a laundry load in the wash basket with a total volume of wash fluid in the wash tub;

operating the drain pump to drain a portion of the total volume of the wash fluid from the wash tub until a remaining portion of the wash fluid remains in the wash tub;

operating the motor to rotate the wash basket at a sensing speed;

obtaining a pressure signal from the pressure sensor; analyzing the pressure signal to identify an out of balance condition; and

implementing a responsive action in response to identifying the out of balance condition.

14. The method of claim **13**, wherein analyzing the pressure signal to identify the out of balance condition comprises:

identifying a maximum pressure and a minimum pressure from the pressure signal;

determining an oscillation amplitude of the pressure signal from the maximum pressure and the minimum pressure; and

identifying the out of balance condition when the oscillation amplitude exceeds a predetermined amplitude threshold.

15. The method of claim **13**, wherein the sensing speed is lower than an extraction speed where the wash fluid is extracted from the laundry load.

16. The method of claim **13**, wherein the sensing speed is less than 50 revolutions per minute.

17. The method of claim **13**, wherein the analysis to identify the out of balance condition is performed before a spin cycle.

18. The method of claim **13**, further comprising:

analyzing the pressure signal to determine the out of balance condition does not exist; and

performing a spin cycle in response to determining the out of balance condition does not exist.

19. The method of claim **13**, wherein implementing the responsive action comprises:

performing a load redistribution procedure.

20. The method of claim **13**, wherein implementing the responsive action comprises:

providing a user notification of the out of balance condition.

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